

Project Documentation: Golden Bangla Bank Ltd. (GBBL) Management System

Project Overview

- **Project Name:** Golden Bangla Bank Ltd. (GBBL) Management System
- **Objective:** To provide a secure, text-based console application for managing user accounts, vault liquidity, deposits, loans, peer-to-peer (P2P) transfers, and ledger auditing in a streamlined manner.

Technical Specifications & Technologies Used

The application is written in standard **C** and utilizes Windows-specific console management libraries to deliver a modern visual interface and data persistence.

Header File Details

- **<stdio.h>**: Provides standard input and output functions (e.g., `printf`, `scanf`, `getchar`) and file operations for data persistence.
- **<stdlib.h>**: Handles memory management, process termination, and executing shell commands (e.g., `system("cls")` to clear the screen).
- **<string.h>**: Used for string manipulation functions such as `strcpy`, `strcspn`, `strcmp`, and `strstr` to handle account names and ledger records.
- **<windows.h>**: Provides access to the Windows API to manipulate the console, including changing colors (`SetConsoleTextAttribute`) and changing the title (`SetConsoleTitle`).
- **<ctype.h>**: Used for character classification and conversion, such as `tolower()`, to ensure menu interactions are case-insensitive.
- **<time.h>**: Handles the system clock to retrieve local time, enabling the system to create timestamped audit trails.

Architecture & Core Mechanics

The system is designed with a procedural architecture using flat-file storage (`GBBL_Accounts.txt` and `GBBL_Ledger.txt`) to maintain data persistence across executions.

Data Entities

- **Account Structure:** Holds financial and operational variables including the unique account number, account holder name, authentication PIN, balance, loan balance, and state flags (active/deactive, reset status).
- **Transaction Ledger:** Records timestamped transactions tagged with account IDs, action types, amounts, and metadata.

Account Number Generation

- The account number generation is fully automated. The first account created starts at `1001` and subsequent accounts increment by 1. The admin account is preserved at `9999`.

Multi-Word String Reading

- Uses `clearBuffer()` and `scanf("%[^\\n]s")` to allow names containing spaces without creating input errors.

System Work & Data Flow

The system initializes by loading existing data files or creating an administrator file if none exists. It then presents a menu to the user.

[Start Program]

|
v

[loadData() & initAdmin()]

|
+--> Reads GBBL_Accounts.txt / Initializes 9999 Admin Account
|
v

[Display Main Menu]

|
+--> [1] Secure Login -----> [Admin Dashboard] or [User Dashboard]
+--> [2] Open Account -----> [Creates New Account with Auto ID]
+--> [3] System Exit -----> [Save Data & Terminate]

[Main Application Startup]

|
+--> [1] Secure Login
| |
| | +--> Admin Dashboard (ID: 9999)
| | +--> 1. Deposit Requests
| | +--> 2. User Database
| | +--> 3. Add Vault Cash
| | +--> 4. Global Ledger
| | +--> 5. Account Stats
| | +--> 6. Pin Reset Requests
| | +--> 7. Loan Requests
| | +--> User Dashboard
| | +--> 1. Deposit Request
| | +--> 2. Withdraw Money
| | +--> 3. Peer Transfer
| | +--> 4. Transaction History
| | +--> 5. Change PIN
| | +--> 6. Change Name
| | +--> 7. Forget Password
| | +--> 8. Request Money
| | +--> 9. Money Requests
| | +--> A. Loan Management
| | +--> V. View Profile
|
+--> [2] Open Account
+--> [3] System Exit

Comprehensive Menu & Submenu Specifications

The system is split into multiple dashboards. Each menu item performs a specific function.

1. Main Authentication Menu

This module processes system initialization, initial ledger reads from the ledger, and session assignment.

Selection	Action	Description
[1]	Secure Login	Prompts for an Account ID and PIN. Automatically routes users to the Admin Dashboard (for the 9999 account) or the User Dashboard (for customer accounts).
[2]	Open Account	Opens an account by prompting for a user name and 4-digit PIN. Assigns an automated, sequential account number beginning at 1001.
[3]	System Exit	Writes updated states to file, executes an animated shutdown, and terminates the application.

2. User Services Dashboard

Accessible to standard customers, this dashboard manages daily transactions, profiles, loans, and peer requests.

Submenu Operations

- **[1] Deposit Request:** Allows the user to enter an amount to be deposited, which is added to a pending approval queue and verified by the admin.
- **[2] Withdraw Money:** Processes cash withdrawals directly from the user's balance, provided there are sufficient funds.
- **[3] Peer Transfer:** Transfers funds directly to another account by inputting the recipient's Account ID and the amount.
- **[4] Transaction History:** Displays a personal ledger of historical transactions associated with the account.
- **[5] Change PIN:** Authenticates the old PIN before updating to a new user-defined value.
- **[6] Change Name:** Updates the account holder's name using an input function that supports multi-word strings.
- **[7] Forget Password:** Flags the account in the administrator database for a PIN reset to 1111.
- **[8] Request Money:** Sends a request signal and amount to a target user's account.
- **[9] Money Requests:** Displays an inbox of pending peer transfer requests, which the user can accept or reject.
- **[A] Loan Management:** Opens the sub-menu for taking out or paying off a loan.
- **[V] View Profile:** Displays user details including name, account number, balance, and account status.

Loan Management Submenu (A)

- **How it works:** Allows users to check their current loan balance and choose between two operations:
 - **Take Loan:** The user inputs an amount, creating a pending request that goes to the admin queue.

- **Pay Loan:** The user can repay outstanding balances from their available funds. The amount is subtracted from the loan balance and credited back to the vault.

3. Admin Command Center

Accessible via the **9999** Administrator account. It provides control over the platform's liquidity and operational requests.

Submenu Operations

- **[1] Deposit Requests:** Views pending deposit requests from customers and either credits the amount or rejects the request.
- **[2] User Database:** Displays all user accounts, toggles active/inactive status, or purges records.
- **[3] Add Vault Cash:** Increases total vault liquidity.
- **[4] Global Ledger:** Displays all logged events across the system.
- **[5] Account Stats:** Displays the total market circulation and the number of active accounts.
- **[6] Pin Reset Requests:** Views requests for PIN resets and processes them by resetting the user's PIN to **1111**.
- **[7] Loan Requests:** Reviews and processes customer loan applications.

User Management Submenu (2)

- **How it works:** Displays account information (such as balances and loan balances) and lets the admin delete accounts or toggle their active status. The admin account (**9999**) is protected from deletion and modification.

Money Requests Submenu (9 in User Menu)

- **How it works:** Displays incoming requests from peers. The user can either accept the request (which checks for sufficient balance before transferring funds) or reject the request.

Detailed Application Startup Guide

Follow these steps to compile and run the system:

1. **Prerequisites:** Ensure you have a standard C compiler such as **gcc** (MinGW) installed on your Windows system.
2. **Setup:** Save the provided C code inside a file named **main.c**.
3. **Compilation:** Open your terminal or command prompt, navigate to your working directory, and run the following compilation command:
4. Bash

```
gcc main.c -o GBBL_System.exe
```

- 5.
- 6.
7. **Execution:** Execute the compiled program from your terminal or double-click the executable:
8. Bash

```
GBBL_System.exe
```

- 9.
- 10.
11. **System Initialization:** * On the first run, the system creates the **GBBL_Accounts.txt** and **GBBL_Ledger.txt** files automatically.
 - It initializes the default administrative account with the credentials **ID: 9999** and **PIN: 9876**.
 - Open user accounts via option **[2]** on the main screen to begin.

Output Analysis: Golden Bangla Bank Ltd. (GBBL) Management System - UI and Menus

This analysis provides a comprehensive overview of the User Interface (UI), screens, and menu navigation of the system.

```
*****
  GOLDEN BANGLA BANK LTD. | WELCOME TO OUR SYSTEM
*****
Built for the Nation, Driven by Integrity

[1] Secure Login      [2] Open Account
[3] System Exit       [R] Refresh UI

-----
"Empowering your future."
-----

Selection > |
```

```
*****
  GOLDEN BANGLA BANK LTD. | LOGIN PORTAL
*****
Account ID (0: Back | 99: Refresh): 9999
Secure PIN: 9876|
```

```
*****
  GOLDEN BANGLA BANK LTD. | USER DATABASE
*****
ACC      | NAME                | BAL          | LOAN          | STATUS
-----|-----|-----|-----|-----
9999     | GBBL Admin          | 1000005678.00 | 0.00          | ACTIVE
10000    | Sahoss Khan New     | 5700.00       | 0.00          | ACTIVE
10001    | Sangram Khan        | 300.00        | 0.00          | ACTIVE

[1] Delete Account  [2] Toggle Status  [99] Refresh  [0] Back
Action > |
```

GOLDEN BANGLA BANK LTD. | ADMIN COMMAND CENTER

Vault Liquidity: \$1000005678.00

- [1] Deposit Requests

[3] Add Vault Cash

[5] Account Stats

[7] Loan Requests
- [2] User Database

[4] Global Ledger

[6] Pin Reset Requests

[R] Refresh System

"Integrity in every transaction."

[L] Logout | Command > |

GOLDEN BANGLA BANK LTD. | LEDGER HISTORY

[03-05-2026 08:33 PM]	ACC:10000	DEPOSIT	\$10000.00		Approved
[03-05-2026 08:34 PM]	ACC:10000	LOAN_IN	\$2000.00		Loan Approved
[03-05-2026 08:36 PM]	ACC:10001	SECURITY	\$0.00		Admin Approved Reset to 1111
[03-05-2026 08:38 PM]	ACC:10000	REQ_SENT	\$100.00		Paid Req to Sangram Khan
[03-05-2026 08:38 PM]	ACC:10001	REQ_RCVD	\$100.00		Req Fulfilled by Sahoss Khan New
[03-05-2026 08:40 PM]	ACC:10000	W-DRAW	\$2000.00		Cash Withdrawal

"Transparency is our commitment."
[0] Back | [99] Refresh

GOLDEN BANGLA BANK LTD. | APPROVAL CENTER

No pending deposit requests.

"System Balanced."
[0] Back | [99] Refresh

GOLDEN BANGLA BANK LTD. | PIN RESET REQUESTS

ACC	NAME	STATUS
-----	------	--------

No pending reset requests found.

"System Secure."

[0] Back | [99] Refresh

GOLDEN BANGLA BANK LTD. | LOAN REQUESTS CENTER

No pending loan requests.

"Stability maintained."

[0] Back | [99] Refresh

GOLDEN BANGLA BANK LTD. | OPEN ACCOUNT

Full Name (0: Back | 99: Refresh): Zahin

Set 4-Digit PIN: 5566

Issuing Account...

ID: 10002

"Remember your credentials"

[0] Back | [99] Refresh

GOLDEN BANGLA BANK LTD. | USER SERVICES

Welcome, Sahoss Khan New

Acc: 10000 | Bal: \$5700.00 | Loan: \$0.00

- | | |
|---------------------|-------------------------|
| [1] Deposit Request | [2] Withdraw Money |
| [3] Peer Transfer | [4] Transaction History |
| [5] Change PIN | [6] Change Name |
| [7] Forget Password | [8] Request Money |
| [9] Money Requests | [A] Loan Management |
| [V] View Profile | [R] Refresh Page |

"Your prosperity, our priority."

[L] Logout | Selection > |

GOLDEN BANGLA BANK LTD. | LOAN SERVICES

Your Loan Balance: \$0.00

[1] Take Loan [2] Pay Loan [R] Refresh [0] Back
Selection > |

GOLDEN BANGLA BANK LTD. | MONEY REQUESTS BOX

ID	REQUESTER	AMOUNT
----	-----------	--------

No incoming requests.

"Peace of mind."

[0] Back | [99] Refresh

```
*****
GOLDEN BANGLA BANK LTD. | MY PROFILE
*****
Name:          Sahoss Khan New
Account No:    10000
Current Balance: $5700.00
Account Status: Active

-----
"Personal Banking Details"
[0] Back | [99] Refresh
-----
```

```
*****
GOLDEN BANGLA BANK LTD. | LEDGER HISTORY
*****
[03-05-2026 08:33 PM] ACC:10000 | DEPOSIT | $10000.00 | Approved
[03-05-2026 08:34 PM] ACC:10000 | LOAN_IN | $2000.00 | Loan Approved
[03-05-2026 08:38 PM] ACC:10000 | REQ_SENT | $100.00 | Paid Req to Sangram Khan
[03-05-2026 08:40 PM] ACC:10000 | W-DRAW | $2000.00 | Cash Withdrawal

-----
"Transparency is our commitment."
[0] Back | [99] Refresh
-----
```

Conclusion: Golden Bangla Bank Ltd. (GBBL) Management System

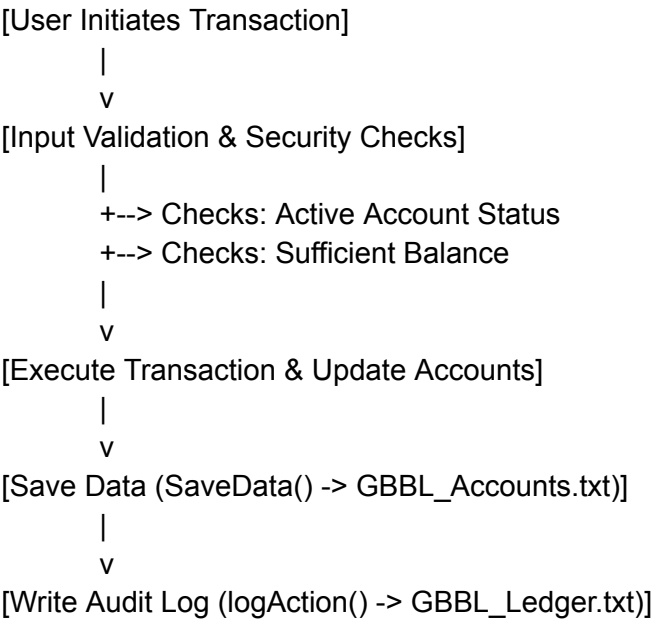
The Golden Bangla Bank Ltd. (GBBL) Management System successfully provides a complete console-based financial management environment. By combining an accessible user interface with secure backend controls, the system ensures reliable operations, accounting accuracy, and data persistence across sessions.

Technical & Operational Achievements

- **Automated Data Persistence:** The system creates and updates the `GBBL_Accounts.txt` and `GBBL_Ledger.txt` flat files after every key operation to prevent data loss.
- **Audit Trail & Ledger Auditing:** Every financial transaction—including deposits, withdrawals, loans, and peer-to-peer transfers—is logged with a timestamp in the global ledger. This ensures transparency and provides a reliable audit trail.
- **Streamlined Authentication & Admin Control:** All requests, such as deposits, loans, and PIN resets, are routed to a central administrative dashboard (ID `9999`) for secure verification.
- **User Security Features:** Includes automated sequential account generation, case-insensitive menu navigation, and account deactivation to prevent unauthorized access.

Architectural Workflow and Security Flow

The workflow below illustrates the steps taken by the system during critical transactions, from initial user requests to audit log generation.



Summary of Core Modules

Module	Key Features & Processes
Authentication Module	Routes users based on their credentials, automatically distinguishing between the administrator and standard customers.
Ledger & Audit Module	Records timestamped entries (including hours, minutes, AM/PM, and amounts) to ensure a complete audit trail.
Loan Management	Handles loan applications, updates the vault balance, and tracks outstanding customer loans.
Peer Services (P2P)	Allows secure transfers directly between user accounts or through incoming/outgoing money requests.

Future Enhancements

- **Database Encryption:** Integrating basic encryption algorithms to increase the security of sensitive data like PINs.
- **Extended Multi-Threading:** Adding a timeout for user sessions to improve security.
- **Dynamic Memory Management:** Replacing the static array limits (**MAX_ACCOUNTS** and **MAX_REQUESTS**) with dynamic arrays to accommodate a growing number of users.

